

Assignment:- 2

Name: Kavshal Kumar Raj

Branch: B.Tech CSE AI & ML

PAN: 240205241023

Semester: IV

Subject: TOC

- ① Prove that the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular using the pumping lemma.

Pr

S1: Assume the expression on regular and accept

String $w = aabb$

No

using pumping lemma

$S_2 = p \mid \forall i \geq 0$ so, $w = a \mid a \mid b b$
 $x \mid x \mid z$

$\therefore x \geq 0$

$S_3 = |xy| \leq n \therefore 3 \leq 4$

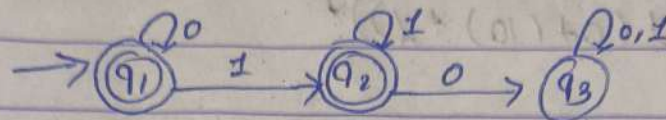
$S_4: x y^i z \in L \forall i \geq 0$

for $i=0$

$w = a(a)^0 b b$
 $= abb \notin L$

Hence this is not a regular language Proved.

② Construct the regular expression for the given DFA using Arden's Theorem.



Solution,

$$q_1 = q_1 0 + \epsilon \quad \text{--- (I)}$$

$$q_2 = q_1 1 + q_2 1 + \epsilon \quad \text{--- (II)}$$

$\therefore q_2$ is also final

$$q_3 = q_3 0 + q_3 1 + q_2 0 \quad \text{--- (III)}$$

Ans

Qn --- (I)

$$q_1 = q_1 0 + \epsilon$$

$$q_1 = \epsilon + q_1 0$$

$$\therefore q_1 = 0^*$$

Put q_1 in q_2 eqn we get

$$q_2 = 0^* 1 + q_2 1 + \epsilon$$

$$q_2 = q_2 1 + (0^* 1 + \epsilon)$$

$$q_2 = (0^* 1 + \epsilon) + q_2 1 \quad \therefore A = Q + RP$$

$$\therefore q_2 = (0^* 1 + \epsilon) 1^*$$

$\therefore$ Regular expression for two final states:

$$\therefore L = q_1 + q_2$$

$$= 0^* + (0^* 1 + \epsilon) 1^*$$

$$= 0^* 1^*$$

Name: Kavushal Kumar Raj

PRN: 240205241023

3) Prove that:

$$(1 + (00)^* 1) + (1 + (00)^* 1) (0 + (10)^* 1)^* (0 + (10)^* 1) \\ = 0^* (1 (0 + (10)^* 1))^*$$

Solution,

$$= (1 + (00)^* 1) [\epsilon + (0 + (10)^* 1)^* (0 + (10)^* 1)]$$

$$= (1 + (00)^* 1) (0 + (10)^* 1)^* \quad \because \epsilon + R^* R = R^*$$

$$= 1 (\epsilon + (00)^*) (0 + (10)^* 1)^*$$

$$= 1 (00)^* (0 + (10)^* 1)^* \quad \because \epsilon + R^* = R^*$$

$$= 1 (0)^* (0 + (10)^* 1)^* \quad \because (R^*)^* = R^*$$

$$= 0^* (1 (0 + (10)^* 1))^* \quad \because PQ^* = (PQ)^*$$

Proved,

4) Prove that the language $L = \{a^n b^n c^n \mid n \geq 0\}$ is not a regular language using Pumping Lemma.

Solution,

Sol: Assume the above language is regular first and accept string $\therefore w = aa \cdot bb \cdot cc$

Name: Koushal Kumar Ray

PRN: 240205241023

$$S2: \exists \geq 0 \quad \therefore w = \underset{x}{aa} \underset{y}{b} \underset{z}{b} cc$$

$$\therefore \exists \geq 0 =$$

$$S3: (xy) \leq n \quad \therefore 3 \leq 6$$

$$S4: xy^iz \in L \quad \forall i \geq 0$$

Now
For $i = 0$

$$\therefore w = aa(b)^0b cc \\ = aabcc \notin L$$

Hence we proved that the above language is not regular.

Sal
18/12/22
(A+)